

DARE COUNTY REGIONAL, NC, USA

WMO: 723046

Lat: 35.917N

Lon: 75.700W

Elev: 13

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 13766

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.4	28.5	7.0	7.9	29.4	11.7	10.0	33.4	26.9	52.9	24.4	48.2	8.4	340	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	91.1	78.4	90.0	78.5	87.9	77.3	81.7	87.6	80.3	86.0	78.8	84.4	9.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	158.8	85.8	78.8	149.7	84.0	77.1	141.2	82.8	45.2	88.2	43.5	86.1	42.4	84.9	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	20.9	19.0	22.4	95.6	4.7	2.8	19.0	97.6	16.3	99.2	13.6	100.8	10.2	102.8		
			18.5	83.4	4.2	1.8	15.5	84.7	13.0	85.8	10.7	86.8	7.6	88.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	43.7	45.7	50.8	60.2	68.6	76.6	80.3	79.4	74.8	65.6	55.4	48.8
	DBStd	14.72	9.45	8.08	8.48	7.77	6.83	5.32	4.19	3.74	4.41	7.07	7.54	8.28
	HDD50	672	242	166	97	8	0	0	0	0	0	0	32	127
	HDD65	2760	661	542	443	184	45	1	0	0	2	80	297	505
	CDD50	5263	46	45	122	315	575	797	941	912	743	483	195	89
	CDD65	1875	0	0	4	41	155	348	476	447	295	98	10	1
	CDH74	15177	0	0	10	134	788	2901	4903	4236	1821	372	12	0
	CDH80	4239	0	0	0	11	142	832	1686	1236	296	36	0	0
Wind	WSAvg	9.2	9.0	9.5	10.4	11.1	10.1	9.1	9.0	8.2	9.2	8.4	8.5	8.5
Precipitation	PrecAvg	48.5	3.9	3.1	3.7	2.7	3.2	5.0	5.7	6.2	5.0	3.5	2.5	3.3
	PrecMax	69.6	7.4	6.5	7.8	7.4	7.1	9.4	10.4	11.7	10.2	8.2	6.9	5.9
	PrecMin	26.9	1.1	0.9	0.8	0.6	0.0	1.0	1.0	3.0	0.7	0.4	0.0	0.7
	PrecStd	9.7	1.9	1.6	1.9	2.0	2.1	2.2	2.4	2.5	2.5	2.1	1.8	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.5	70.1	74.9	81.4	88.2	93.1	96.7	93.2	89.6	83.2	75.3	71.3
		MCWB	61.7	61.6	62.6	67.4	73.9	76.9	79.5	80.3	76.4	73.9	68.2	64.9
	2%	DB	64.1	64.4	70.5	78.0	83.8	90.3	91.4	90.4	85.3	80.8	72.5	67.3
		MCWB	59.6	58.2	61.4	66.7	72.5	76.5	79.0	78.9	76.3	73.0	66.0	63.9
	5%	DB	61.1	61.4	66.6	74.7	80.9	87.8	90.2	88.3	82.6	78.6	69.8	64.1
		MCWB	56.9	56.1	60.6	65.7	71.1	75.6	79.4	78.1	74.7	71.4	64.3	61.0
	10%	DB	57.2	57.6	63.9	72.2	78.6	84.2	88.2	85.9	81.4	75.3	66.4	61.2
		MCWB	54.0	53.2	58.5	64.9	70.1	75.2	78.4	77.2	74.3	69.6	62.5	57.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.7	63.7	66.4	72.5	76.7	80.8	84.2	83.6	80.2	76.9	71.1	67.2
		MCDB	66.0	67.7	71.5	78.1	84.2	87.5	90.5	89.5	84.5	81.5	74.0	68.8
	2%	WB	61.0	60.6	63.9	69.4	74.3	78.8	82.4	81.3	77.8	74.9	67.4	64.5
		MCDB	63.7	63.3	67.5	74.3	80.8	85.4	88.8	86.3	82.3	79.3	69.8	66.2
	5%	WB	58.1	57.7	61.9	67.6	72.9	77.5	80.8	79.9	76.9	73.3	65.3	61.9
		MCDB	60.9	61.0	66.0	72.8	78.9	84.6	87.2	84.8	81.4	77.2	68.2	64.1
	10%	WB	54.9	54.3	59.2	65.9	71.4	76.3	79.4	78.5	75.9	71.0	63.1	58.7
		MCDB	57.2	57.2	63.6	70.9	76.7	83.0	85.4	83.6	80.4	74.6	65.7	61.1
Mean Daily Temperature Range	5% DB	MDBR	13.5	14.2	14.2	14.4	13.1	11.9	11.4	10.8	10.1	12.2	13.2	12.9
		MCDBR	17.0	18.3	17.6	16.9	15.4	14.9	14.2	13.2	12.4	12.6	15.5	14.8
		MCWBR	13.6	13.7	11.0	9.3	7.9	7.2	7.2	6.5	6.4	7.3	11.1	12.4
	5% WB	MCDBR	15.4	17.2	15.7	15.0	13.3	13.0	13.0	12.0	10.3	9.8	13.3	13.9
		MCWBR	13.9	14.8	11.9	9.7	7.7	7.1	7.6	6.7	6.2	7.3	11.3	13.0
Clear-Sky Solar Irradiance	taub	0.324	0.331	0.354	0.398	0.441	0.486	0.513	0.501	0.419	0.368	0.337	0.329	
	taud	2.535	2.510	2.468	2.355	2.274	2.198	2.130	2.155	2.386	2.512	2.545	2.548	
	Ebn at Noon	278	289	290	281	269	256	248	249	266	272	271	269	
	Edh at Noon	25	29	33	39	43	46	49	47	36	29	25	24	
All-Sky Solar Radiation	RadAvg	754	1003	1345	1716	1929	2041	1923	1752	1451	1167	852	681	
	RadStd	52	101	101	125	138	107	98	116	92	115	60	53	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+295	+202

Nomenclature: See separate page